# A cubic source for the vorticity maximum, and a recorded gap in the periodic continuation criterion

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This document does not claim a solution of the Navier–Stokes Millennium Problem.

**Abstract**

For a smooth divergence-free solution of the unforced periodic Navier–Stokes equations, let $M(t) = \|\omega(t)\|_{L^\infty}$. At a spatial maximum of $|\omega|$, transport vanishes and viscosity only lowers $M$, so $M' \le \omega \cdot S\omega / M$. The 7 October version claimed $\omega \cdot S\omega \le C_{\mathrm{CZ}} M^3$ from Calderón–Zygmund theory. That step is *not* proved: it needs $\|\nabla u\|_{L^\infty} \le C\|\omega\|_{L^\infty}$, which Calderón–Zygmund theory does not give. This corrected version replaces it by the classical log-corrected bound, and separately records the uncorrected inequality as a named hypothesis. The comparison ODEs, the ceiling under a quartic sink $\kappa M^4$, and the implication from that sink to a finite Beale–Kato–Majda integral are checked in Lean. The sink itself remains open. Numerics at the vorticity maximum are reported and show no sign of a sink.

## 1 Setting

Let $u$ be a smooth divergence-free solution of the unforced periodic Navier–Stokes equations on $\mathbb{T}^3$, with viscosity $\nu > 0$, and let $\omega = \nabla \times u$, $S = \frac{1}{2}(\nabla u + \nabla u^\top)$. Write $M(t) = \|\omega(t)\|_{L^\infty}$. The vorticity equation is

$$\partial_t \omega + (u \cdot \nabla)\omega = (\omega \cdot \nabla)u + \nu \Delta \omega, \qquad \omega \cdot (\omega \cdot \nabla)u = \omega \cdot S\omega.$$

## 2 The pairing at the maximum, and the corrected source bound

**Proved (classical).** At a spatial maximum $x$ of $|\omega|$, $\omega \cdot (u \cdot \nabla)\omega = \frac{1}{2} u \cdot \nabla |\omega|^2 = 0$, and $\omega \cdot \Delta \omega = \frac{1}{2}\Delta|\omega|^2 - |\nabla \omega|^2 \le 0$. Hence, in the sense of upper Dini derivatives,

$$M\, M' \le \omega(x) \cdot S(x)\omega(x) \le \|S\|_{L^\infty} M^2. \tag{1}$$

**The inconsistency in the 7 October version.** That version stated that Biot–Savart is a Calderón–Zygmund operator bounded from $L^\infty$ to BMO and *not* on $L^\infty$, and in the next line used $\omega \cdot S\omega \le C_{\mathrm{CZ}} M^3$. The second statement requires $\|S\|_{L^\infty} \le C_{\mathrm{CZ}} \|\omega\|_{L^\infty}$, which is exactly the $L^\infty$ bound the first statement denies. The Riesz transforms are unbounded on $L^\infty$, so the bound $M' \le C_{\mathrm{CZ}} M^2$ was not proved and is withdrawn as a proved statement.

**Replacement (a): the log-corrected bound (classical, not formalized here).** The Beale–Kato–Majda / Kato–Ponce estimate gives, for $s > 3/2$,

$$\|\nabla u\|_{L^\infty} \le C\, M\Big(1 + \log^+ \frac{\|\omega\|_{H^s}}{M}\Big) + C\|\omega\|_{L^2},$$

so

$$M' \le CM^2\Big(1 + \log^+ \frac{\|\omega\|_{H^s}}{M}\Big) + C\|\omega\|_{L^2} M. \tag{2}$$

The logarithm involves a higher norm, which is controlled only through the BKM integral itself; (2) is therefore an ingredient of the BKM argument, not a closed ODE for $M$.

**Replacement (b): conditional, under a named hypothesis.**

**Definition 1** (`PeriodicStrainLinfBound`, open)**.** There is $C \ge 0$ with $\|S(t,x)\| \le C\|\omega(t)\|_{L^\infty}$ for all smooth maximal solutions, all $t$ below the lifespan, and all $x$.

This is not known and is likely false as a uniform statement. Under it, $\omega \cdot S\omega \le CM^3$ (Lean: `pairing_le_of_periodicStrainLinfBound`) and therefore $M' \le CM^2$.

## 3 What the ODEs give (proved in Lean)

All statements in this section are abstract real-analysis lemmas in `CubicSource.lean`, with no `sorry` and only the standard axioms.

**Proposition 1** (`quadratic_comparison`, `lifespan_lower_bound`)**.** *If $M > 0$ is differentiable and $M' \le CM^2$, then $1/M(t) \ge 1/M_0 - Ct$, so $M(t) \le M_0/(1 - CM_0 t)$ for $CM_0 t < 1$.*

This is a lifespan lower bound $T^* \ge 1/(CM_0)$ only. The comparison solution $y_0/(1 - Cy_0 t)$ blows up at $1/(Cy_0)$, so this cannot close BKM.

**Proposition 2** (`cubic_comparison`)**.** *If $C \ge 0$ and $M' \le CM^2(1 + \log M)$, then $1/M(t)^2 \ge 1/M_0^2 - 2Ct$.*

Again only a lifespan lower bound: $\int^\infty dM/(M^2 \log M) < \infty$, so even a self-referential log correction does not prevent finite-time blowup of the comparison.

**Proposition 3** (`fence_ceiling`, `sink_ceiling`)**.** *If $M$ is continuous with right derivative $M' \le M^2(C - \kappa M)$, $\kappa > 0$, then $M \le \max(M_0, C/\kappa)$.*

## 4 The open gap: the quartic sink

**Definition 2** (`PeriodicQuarticVorticitySink`, open, never assumed)**.** For every smooth maximal solution there are $C$ and $\kappa > 0$ such that, below the lifespan, $M$ is continuous with right derivative $M' \le M^2(C - \kappa M)$; equivalently, at the maximum, $\omega \cdot S\omega + \nu\, \omega \cdot \Delta\omega \le CM^3 - \kappa M^4$.

**Theorem 1** (`bkm_bound_of_quarticSink`, proved)**.** *The quartic sink gives $c$ with $M \le c$ and $\int_0^T M\, dt \le cT$ for every $T$ below the lifespan. In particular the BKM integral is finite on $[0, T^*)$ whenever $T^* < \infty$, which by Beale–Kato–Majda would contradict finite lifespan.*

The sink is not derived from the unmodified vorticity equation. It is at least as strong as a uniform bound on the vorticity maximum, i.e. it is the problem restated as an inequality.

## 5 Three false fixes (unchanged)

(i) A formal certificate that expands the assumed rate $M(CM - \kappa M^2)$ into $CM^2 - \kappa M^3$ is ring arithmetic on a hypothesis. (ii) The weight $1 + \kappa M^2$ turns the source into $CM^3 + C\kappa M^5$: a degree-five source, not a sink. (iii) The logarithmic repair needs control of $\|\omega\|_{H^s}$ and is an integral criterion, not a derivation of the sink.

## 6 Numerics at the vorticity maximum (not proofs)

Pseudo-spectral RK4 runs on $\mathbb{T}^3$ (solver of `paper/viscous_balance_tests.py`; script `paper/cubic_source/cubic` random datum peaked at $|k| = 3$, snapshots every 0.05. At $x^* = \arg\max|\omega|$ we record $\sigma = \omega \cdot S\omega$, $\sigma/M^3$, $\|S\|_{L^\infty}/M$, $\nu\,\omega \cdot \Delta\omega/M^2$, and the proxy $\kappa_{\text{eff}} = (C_{\text{run}}M^3 - \sigma)/M^4$ with $C_{\text{run}} = \max \sigma/M^3$.

| $N$ | $\nu$ | seed | $t_{\max}$ | $M$ | $\sigma/M^3$ | slope | $\|S\|_\infty/M$ | min visc$/M^2$ | min $\kappa_{\text{eff}}$ |
|---|---|---|---|---|---|---|---|---|---|
| 64 | 0.0025 | 1 | 8.00 | 9.1–43.0 | -0.068–0.172 | -0.84 | 0.34–0.53 | -0.0705 | 2.2e-03 |
| 64 | 0.005 | 1 | 8.00 | 5.6–29.1 | -0.040–0.141 | -0.23 | 0.37–0.51 | -0.1395 | 1.8e-03 |
| 64 | 0.005 | 2 | 8.00 | 5.8–28.7 | -0.049–0.149 | -0.46 | 0.35–0.61 | -0.1324 | 1.5e-03 |
| 64 | 0.01 | 1 | 8.00 | 2.7–20.2 | 0.017–0.210 | -0.34 | 0.40–0.53 | -0.3205 | 6.2e-03 |
| 96 | 0.0025 | 1 | 8.00 | 10.4–49.7 | -0.033–0.129 | -0.28 | 0.33–0.55 | -0.0829 | 4.3e-04 |
| 96 | 0.005 | 1 | 8.00 | 5.8–30.9 | -0.014–0.139 | -0.13 | 0.36–0.54 | -0.1529 | 1.2e-03 |

**Reading.** (1) $\|S\|_{L^\infty}/M$ stays in about $[0.33, 0.61]$ over the observed range. This is consistent with `PeriodicStrainLinfBound` on these flows, and says nothing about it in general. (2) $\sigma/M^3$ is mostly positive, with a maximum of 0.12–0.21 per run. Fitting $\log(\sigma/M^3)$ against $\log M$ over the positive snapshots gives slopes between $-0.84$ and $-0.13$. So the source is roughly cubic, or somewhat weaker than cubic, at the maximum. (3) $\sigma$ is negative at some snapshots (0–22% of them; vortex compression at the maximum). These snapshots are intermittent: the fraction in the top quartile of $M$ (0–20%) is comparable to the overall fraction, with no systematic trend with $M$. A sink $\kappa M^4$ would require $\sigma/M^3 \le C - \kappa M$ at every large $M$; the data show no such ordering. (4) $\kappa_{\text{eff}}$ is nonnegative only by the choice of $C_{\text{run}}$. Its minimum over the top quartile shrinks toward 0 (down to $4 \times 10^{-4}$ at the finest run), which is the opposite of a uniform $\kappa > 0$. (5) The viscous term at the maximum is negative, of order 10–35 $\nu M^2$: a quadratic, not quartic, sink. *Caveats:* grids $64^3$ and $96^3$, all six runs complete to $t = 8$; peak vorticity only up to about 50; one or two seeds; decaying, unforced flow; snapshots every 0.05; the arg max is taken on the grid. Nothing here bears on behaviour near a putative singularity.

## 7 Record

**Proved:** the pairing (1); the comparison lemmas; the sink ceiling; sink ⇒ finite BKM integral. **Conditional:** $M' \le CM^2$ under `PeriodicStrainLinfBound`; the log-corrected source (2) is classical and not formalized here. **Open:** `PeriodicQuarticVorticitySink`. No theorem in this note proves global regularity for the periodic Navier–Stokes equations.